Check for updates

Received: 3 July 2023 | Revised: 30 January 2024 | Accepted: 11 February 2024

DOI: 10.1111/sms.14586

ORIGINAL ARTICLE

WILEY

Are all hamstring injuries equal? A retrospective analysis of time to return to full training following BAMIC type 'c' and T-junction injuries in professional men's rugby union

Fearghal Kerin[1] | **Stuart O'Flanagan**[2,3] | **Joe Coyle**[3] | **Darragh Curley**[2] | **Garreth Farrell**[2] | **Ulrik McCarthy Persson**[1] | **Giuseppe De Vito**[4] | **Eamonn Delahunt**[1,5]

[1]School of Public Health, Physiotherapy and Sports Science, University College Dublin, Dublin, Ireland

[2]Leinster Rugby, Dublin, Ireland

[3]Sports Medicine Department, UPMC Sports Surgery Clinic, Dublin, Ireland

[4]University of Padova, Padova, Italy

[5]Institute for Sport and Health, University College Dublin, Dublin, Ireland

Correspondence
Fearghal Kerin, School of Public Health, Physiotherapy and Sports Science, University College Dublin, Dublin, Ireland.
Email: fearghal.kerin@ucdconnect.ie

Funding information
Irish Research Council

Abstract

We aimed to determine whether the anatomical location (intramuscular tendon or T-Junction) of hamstring muscle injuries in professional men's rugby union associates with a prolonged time to return to full training and a higher rate of re-injury/subsequent injury. We reviewed the medical records of an Irish professional rugby union club to identify hamstring muscle injuries incurred across five seasons. Clinicians and players were not blinded to MRI results at the time of rehabilitation. A blinded musculoskeletal radiologist re-classified all included injuries ($n = 91$) according to the British Athletics Muscle Injury Classification framework. Players who sustained an injury with intramuscular tendon involvement required a longer time to return to full training compared to players who sustained an injury without intramuscular tendon involvement (78 days vs. 24 days). Players who sustained a biceps femoris injury with T-junction involvement did not require a longer time to return to full training compared to players who sustained a biceps femoris injury without T-junction involvement (29 days vs. 27 days). Injuries with either intramuscular tendon or T-junction involvement were not associated with an increased rate of re-injury/subsequent injury to the same limb (intramuscular tendon involvement – odds ratio = 0.96, T-junction involvement – odds ratio = 1.03). When a hamstring muscle injury involves the intramuscular tendon, the injured player and stakeholders should be made aware that a longer time to return to full training is likely required. T-junction involvement does not alter the expected clinical course of biceps femoris injuries.

KEYWORDS

hamstring muscles (MeSH), rehabilitation (MeSH), soft tissue injuries (MeSH), sports injuries (MeSH)

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.
© 2024 The Authors. *Scandinavian Journal of Medicine & Science In Sports* published by John Wiley & Sons Ltd.

1 | INTRODUCTION

Hamstring muscle injuries (HMIs) are the most prevalent lower limb muscle injury during participation in field sports.[1–4] HMIs are common in rugby union[3,5] and have been described during specific game-related tasks including, running, decelerating, kicking, tackling and rucking.[6] At the 2019 Rugby World Cup, HMIs accounted for 10% of all injuries, and were second only to knee injuries as a cause of injury-related time loss.[7] The prevalence and incidence rate of HMIs is amplified by the frequent occurrence of re-injuries.[8] It has been reported that re-injuries are often more severe.[3,9,10] Re-injury rates of 7%–45% have been reported in rugby union.[3,5,11]

Determining the severity and site of a muscle injury through magnetic resonance imaging (MRI) is often the first step in the management of HMIs in elite sport.[12] However, the benefits of traditional MRI grading frameworks over clinical diagnosis in establishing prognosis following HMIs are unclear.[13] As a result, novel classification frameworks have been established which expand upon previously established methods, particularly in classifying the specific anatomical location of the injury.[14] The British Athletics Muscle Injury Classification (BAMIC)[15] is commonly used in clinical practice and research[12,16–20] (Appendix S1). Injuries are graded from 1 to 4 depending on the size of the injury relative to the cross-sectional area of the muscle. The novel aspect of the BAMIC framework is the description of injury location within the musculotendinous unit. The suffix 'a' denotes a myofascial injury in the peripheral aspect of the muscle, 'b' an injury within the muscle belly, most commonly at the muscle tendon junction (MTJ) and 'c' an injury which extends into the intramuscular tendon. As with the numerical grading, this lettering system is intended to indicate increasing severity.

Several studies across sports have reported time to return to full training and increased re-injury rates following intramuscular tendon disruption (BAMIC 'c').[17,18,21–23] However, the results of these studies are limited by the relatively small numbers of intramuscular tendon injuries ($n = 12$ – $n = 16$) and heterogenous methodologies. Not all studies have reported increased re-injury rates and longer return to play times. van der Made et al.[23] reported only a modest increase in recovery time when clinicians were biased to the presence of intramuscular tendon injury, while a retrospective cohort study of Premier League soccer players did not find that intramuscular tendon injury influenced time to return to play.[19] Wangensteen et al.[14] and van der Made et al.[24] demonstrated no difference in re-injury risk following intramuscular tendon injury. While it has been suggested that differences in the relative demands of sports within these cohorts may explain these inconsistencies,[12] it is necessary to further investigate this across other sports with high prevalence and incidence rates of HMIs, including men's professional rugby union.

In addition to those locations described by the BAMIC framework, other HMIs locations may also be associated with poor outcomes. Entwisle et al.[25] reported that injuries to the distal T-Junction (the confluence of the epimyseal surfaces of the anterolateral long head and posterolateral short head of the biceps femoris) have particularly high re-injury rates – reported at 54%–86%. Several other studies have supported the suggestion that distal injuries may have worse outcomes.[5,18,22] However, Pollock et al.[16] found no difference in outcomes between HMIs that did and did not involve the T-Junction. This disparity in findings around this relatively new area of interest warrants further investigation.

The aims of this retrospective study within a single European professional men's rugby union team were to:

1. Describe the characteristics of HMIs (injury location, BAMIC grade, time to return to full training, and the rate of re-injury/subsequent injury)
2. Analyze the influence of intramuscular tendon injury (BAMIC class 'c') on the time to return to full training and the rate of re-injury/subsequent injury
3. Analyze the influence of T-Junction injury on the time to return to full training and the rate of re-injury/subsequent injury

It was hypothesized that HMIs with intramuscular tendon or T-junction involvement would have a longer time to return to full training and a higher rate of re-injury/subsequent injury.

2 | MATERIALS AND METHODS

2.1 | Design

This study was a retrospective cohort study in a single European professional men's rugby team. Ethical approval was received from the University College Dublin Human Research Ethics Committee. Appendix S2 provides an overview of the overall study procedure.

2.2 | Participants and data collection

Records of time-loss HMIs sustained by Leinster Rugby players across 5 competitive seasons (July 2015–June 2021) were obtained from the club's electronic medical records system (EMRS). The combined senior and academy squads consisted of approximately 60 players per competitive season. All radiologically confirmed acute HMIs

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(confirmed via MRI within 5 days of injury) occurring during that period and that resulted in absence from normal training and match play of more than 7 days were eligible for inclusion.

2.3 | Magnetic resonance imaging (MRI)

Each MRI had been performed on a 3 T MRI scanner at a single imaging center in Dublin, Ireland. A consultant musculoskeletal radiologist (JC) who was experienced in using the BAMIC framework to classify HMIs and in identifying T-Junction injuries, retrospectively reviewed each set of images; he was blinded to the injury outcomes. Location (proximal – above the lower fibers of gluteus maximus, central or distal – below the origin of the short head of biceps femoris), involved muscle, injury site ('a', 'b', 'c' or 'free tendon'), MRI characteristics and injury grading were recorded. Findings were reported on an MRI proforma, which was adapted from Pollock et al.[17] (Appendix S3). The 'free tendon' classification was used for injuries to the proximal or distal section with no muscle fibers attached. T-Junction injuries (at the confluence of the long head and short head of the biceps femoris) were classified as 'Yes/No', using the description previously outlined by Entwisle et al.[25]

2.4 | Rehabilitation

Following injury, all players received acute injury management, physiotherapy, and sports-specific rehabilitation tailored to both their clinical presentation and the sporting context. Players and clinicians were not blinded to MRI findings at the time of injury. This was overseen by the club's physiotherapy and sports medicine team who provided supervised, criteria-based rehabilitation specific to player's presentation. Progressive milestones—including pain-free palpation and clinical testing through range in the early stage, strength testing (isometric, isokinetic and eccentric hip, and knee dominant exercises) in the middle stage and finally running and training replication drills monitored by GPS, and exposure to tackling and sprinting in the final stage—were cleared before the players were cleared to return to full training.

2.5 | Time to return to full training

The EMRS were consulted to evaluate the time to return to full training for each player included in the study. This outcome metric was deemed most appropriate as players may not have been selected for matches despite being deemed match fit. This analysis was performed by two physiotherapists (FK and DC) who independently reviewed the EMRS and determined the day when each player had been cleared to return to unrestricted training, including sprinting and full contact sport-specific activities. For injuries suffered toward the end of a season, and which extended into the off-season, time to return to full training was calculated from the time they achieved their prescribed rehabilitation criteria. HMIs not included in this study included those whereby the player: (1) sustained a concurrent injury which affected his rehabilitation, (2) had delayed or revision surgery, (3) was not available for follow up 12 months post-injury (Appendix S4).

2.6 | Re-injury/subsequent injury

The EMRS were consulted to identify further HMIs suffered by players in the 12 months following time to return to full training after their index injury. These injuries were classified in the same manner described earlier (Magnetic resonance imaging section). An additional three subclassifications were used to describe these injuries; (1) re-injury (injury to same muscle), (2) subsequent injury to the same leg, and (3) subsequent injury to the other leg. They were also further defined as: (1) early stage (<3 months), (2) medium stage (3–6 months), or (3) late stage (6–12 months) re-injuries/subsequent injuries, depending on when they were suffered relevant to the time to return to full training after the index injury. In cases where a player had more than one subsequent injury, only his first subsequent injury was included in this study. Re-injuries/subsequent injuries that were suffered during rehabilitation and before time to return to full training were considered a prolongation of the index injury and were not recorded as a re-injury/subsequent injury.[26]

2.7 | Equity, diversity, and inclusion

This study focused exclusively on male professional rugby union players. We acknowledge that our author list does not reflect the diversity of the sport and exercise medicine/rehabilitation community.

2.8 | Statistical analysis

We used descriptive statistics to describe the classifications and outcomes for the HMIs included in this study. Additionally, a one-way ANOVA was performed to determine if there was a significant difference in the average time to return to training across the BAMIC classifications

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of injury. Tukey's test was used to correct for multiple comparisons. The dependent variable was time to return to full training, with the independent variables being BAMIC classification of injury (1a, 1b, 1c, 2a, 2b, 2c, 3c, 4c).

To evaluate the influence of intramuscular tendon injury (BAMIC 'c') on time to return to full training, we performed a Kaplan Meier analysis. This involved constructing survival curves for each group. These curves illustrate the proportion of injuries remaining having resumed full training over time. We aggregated 'a' and 'b' injuries to create a combined curve, providing a reference for cases without intramuscular tendon involvement. Simultaneously, the 'c' injuries were aggregated to form a separate curve, representing cases with intramuscular tendon involvement. By comparing these curves, we aimed to discern any notable differences in the time to return to full training between injuries with and without intramuscular tendon involvement. This approach allows for a nuanced understanding of the influence of intramuscular tendon injury on the observed outcomes.

To evaluate the influence of intramuscular tendon injury (BAMIC 'c') on the rate of re-injury/subsequent injury we performed a Fisher exact test. We also calculated the associated odds ratio. For this analysis, all 'a' and 'b' injuries (no intramuscular tendon involvement) were aggregated, while all 'c' injuries (intramuscular tendon involvement) were aggregated.

To evaluate the influence of T-junction injury on time to return to full training, we again performed a Kaplan Meier analysis. For this analysis, we grouped data as follows: biceps femoris T-junction involvement vs. biceps femoris no T-junction involvement. To evaluate the influence of T-junction injury on the rate of re-injury/subsequent injury we performed a Fisher exact test. We also calculated the associated odds ratio. For this analysis, we grouped data as follows: biceps femoris T-junction involvement vs. biceps femoris no T-junction involvement. Our statistical analysis are consistent with the CHAMP statement.[27]

All statistical analyses were performed using GraphPad Prism version 9.0 for MacOS, GraphPad Software, San Diego, California USA.

3 | RESULTS

3.1 | Participants

One hundred and forty three injuries registered on the EMRS over 5 competitive seasons were assessed for eligibility (Appendix S4). Eighteen injuries were available for full training in less than 7 days, and in 22 cases no MRI investigation was performed. Two injuries were classified as free tendon injuries and were excluded from further analysis. One case was excluded as the MRI located the injury to the quadratus femoris muscle. In one case, consensus on time to return to full training could not be reached by the investigators. One player left the club within 12 months of his injury and thus, could not be followed up. In two cases, the players had other subsequent injuries during rehabilitation that extended their time to return to full training. Eleven cases that were managed with surgery were included. However, one player had delayed surgery after trialing rehabilitation, while another had a revised surgery; both were excluded. Following this process, 91 HMIs suffered by 46 players over 5 competitive seasons (see Appendix S3) were included for final analysis. Sixteen players (18%) experienced ≥2 separate HMIs.

3.2 | Description of injuries

Table 1 describes the characteristics of the HMIs. Twenty six (29%) injuries were proximal, while thirty seven (41%) injuries were distal. The most frequently injured muscle in isolation was biceps femoris long head (53%), followed by semitendinosus (22%). Grades 1, 2, 3, and 4 accounted for 27, 49, 13, and 2 HMIs respectively (Table 2). The most common class was BAMIC 'b' ($n = 52$, 57%), followed by BAMIC 'c' ($n = 26$, 29%), and BAMIC 'a' ($n = 13$, 14%). Sixteen of the 26 BAMIC 'c' type injuries involved some ($n = 8$) or complete ($n = 8$) loss of tension. Eighteen of 30 distal biceps femoris injuries involved disruption of the T-Junction. The average time to return to full training following HMI was 39.5 days (SD = 38 days; 95% CI = 31–47 days) (Table 3). Ten per cent of HMIs re-injured early (<3 months), with a total of 13% re-injuring within 12 months of returning to full training (Table 3 – time to return to full training and re-injury by Location). In addition, a further 10% of HMIs had a subsequent early injury (<3 months) to the same leg, with a total of 15% having a subsequent injury to the same leg within 12 months of returning to full training. Seven per cent suffered a subsequent HMI to the other leg within 3 months of returning to full training, with a total of 18% suffering a subsequent HMI to the other leg within 12 months of returning to full training.

A significant difference was observed for the time to return to full training across the BAMIC grades of injury ($p < 0.0001$). These data are presented in Figure 1.

3.3 | Intramuscular tendon injury

Results of the Log-rank (Mantel-Cox) test determined that the median time to return to full training was 18 days

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Characteristics of the study participants and hamstring muscle injuries.

Variable	Number	Per cent
Study participants, $n=46$		
Mean age* in years (SD: range)	25.5 (4.5; 19–36)	n/a
Players who suffered 2 or more separate hamstring injuries	16	35%
Mean number of injuries (median, range), $n=16$	3.8 (3.5; 2–7)	n/a
Injuries		
Total number of injuries in the sample of 46 players	91	n/a
Mean time to return to full training in days (SD; range)	39.5 (38.2; 8–180)	n/a
Side of injury		
Right	46	51%
Left	47	51%
Location of the injury		
Proximal	26	29%
Central	28	31%
Distal	37	41%
Muscle injured		
Biceps femoris (long head)	48	53%
Semitendinosus	20	22%
Semimembranosus	17	19%
Biceps (short head)	4	4%
Multiple (biceps femoris long head and semitendinosus)	2	2%
Subsequent injury (acute hamstring injury to any muscle of either leg within 12 months)		
Yes	42	46%
No	49	54%
Re-injury to same muscle	13	14%
Subsequent injury to other hamstring muscle of same leg	14	15%
Subsequent injury to any hamstring muscle of other leg	16	18%
Early	24	26%
Middle	10	11%
Late	8	9%
Surgery		
Yes	11	12%
No	80	88%
Intramuscular tendon involvement		
Yes	26	29%
No	65	71%
T-Junction involvement		
Yes	18	20%
No	73	80%
Of distal Biceps Femoris injuries ($n=30$)	23%	60%

Abbreviations: SD: standard deviation, n/a: not applicable.

*Age at the time of injury.

(inter-quartile range = 14–24 days) for the HMIs without intramuscular tendon involvement, and 81 days (inter-quartile range = 38–112 days) for the HMIs with intramuscular tendon involvement. This difference was statistically significant ($p<0.0001$; hazard ratio (log rank) for the HMIs that had no intramuscular tendon involvement = 2.99, 95% confidence interval = 1.98 to 4.51) (Figure 2).

Results of the Fischer exact test determined that there was no statistically significant difference in the rate of re-injury/subsequent injury for HMIs with or without intramuscular tendon involvement. The odds ratio for subsequent injury (for HMIs with intramuscular tendon involvement) was 0.96 (95% CI = 0.37 to 2.64).

3.4 | T-junction injury

Results of the Log-rank (Mantel-Cox) test determined that the median time to return to full training was 18 days (inter-quartile range = 12–30 days) for biceps femoris injuries with no T-junction involvement and 24 days (inter-quartile range = 10–44 days) for biceps femoris injuries with T-junction involvement. This difference was not statistically significant $p<0.0001$; hazard ratio (log rank) for the biceps femoris injuries that had no T-junction involvement = 1.08, 95% confidence interval = 0.59 to 1.94 (Figure 3).

Results of the Fischer exact test determined that there was no statistically significant difference in the rate of re-injury/subsequent injury for biceps femoris injuries with or without T-junction involvement. The odds ratio for subsequent injury (for biceps femoris injuries with T-junction involvement) was 1.03 (95% CI = 0.30 to 3.99).

4 | DISCUSSION

It has been suggested that the site of injury is important in the management of HMIs.[28] This retrospective analysis

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 Primary injury location and muscle injury grading using the BAMIC.

BAMIC Classification	Biceps femoris long head	Biceps femoris short head	Semitendinosus	Semimembranosis	Multiple muscles (BF/ST)	Total
1a	2	0	3	1	0	6
1b	12	0	4	4	1	21
2a	0	1	3	3	0	7
2b	18	2	8	2	1	31
2c	7	1	1	2	0	11
3a	0	0	0	0	0	0
3b	0	0	0	0	0	0
3c	8	0	0	5	0	13
4c	1	0	1	0	0	2
	48	4	20	17	2	91

Abbreviations: BAMIC: British Athletics Muscle Injury Classification; BF: biceps femoris; semitendinosus: ST.

of 91 HMIs suffered by 46 players over 5 competitive seasons is the first to describe HMI grades and location using the BAMIC framework, time to return to full training and re-injury/subsequent injury rates in professional rugby union.

4.1 | Injury description

The most frequently injured muscle was biceps femoris long head (55% of all injuries). This is a lower percentage than that previously reported in rugby union by Bourne et al.[11] (80%) and Kenneally-Dabrowski et al.[5] (73%). Our findings are similar to those reported in elite-level English football, where biceps femoris injuries have been shown to comprise 48–66% of all HMIs.[18,19] Twenty two percent of the HMIs in our study involved the semitendinosus muscle. Other groups have reported that semitendinosus injury accounts for 4%–14% of HMIs in athletics and football[16,17,19]

Thirteen of the injuries included in this study were classified as grade 3 (14%), and two were classified as grade 4 (2%). This is a higher percentage of these grades of injuries than what has been previously reported in football (0%–6% of HMIs), and consistent with the proportion of grade 3 injuries that have been reported to (13%–18%).[18,19,24] Our results regarding the percentage of intramuscular tendon injuries (24%) is similar to those reported in published studies (17%–39%).[18,19,21] As we excluded injuries with a time to return to full training of less than 7 days and those that had not underwent MRI, we may underrepresent lower grade injuries that quickly returned to full training. Similarly, it is possible that there are intramuscular tendon or T-Junction injuries with very short time to return to full training that were not considered due to the methodology of this study.

Our results describe a higher percentage (41%) of distal HMIs than that reported by Pollock et al.[17] (28%), but less than that previously reported in rugby union[5] (60%), although the latter characterized injuries as either proximal or distal, not central as in the present study. Twenty per cent of HMIs in our study involved the distal musculotendinous T-Junction of biceps femoris. This is higher than the percentage previously described in a group primarily comprised of Australian Rules football players (6%).[25] No other authors have yet to describe the percentage of HMIs that extend in to the T-Junction.

Almost half (45%) of the players included in our study suffered a subsequent HMI to either leg within 12 months of returning to full training. While 12% of subsequent HMIs were re-injuries to the same muscle, other studies that do not evaluate other muscle injuries to the same leg may fail to register potentially related subsequent injuries (15%). Our results suggest that the burden of subsequent HMI is not just caused by immature healing at the original injury site, but potential sensorimotor impairments which heighten the risk of injury to other lower limb muscles.[29] Fifty seven per cent of subsequent injuries were suffered during the first 3 months following return to full training. While this is a large percentage and warrants consideration in the mitigation of subsequent injury risk, some similar studies to ours only report re-injuries up to 3 months following return to play, and thus have not fully quantified the medium- and long-term sequelae of HMIs.[16–18]

The mean time to return to full training following HMI in our cohort was 39.5 days (95% CI = 31–47 days). This is longer compared to previously reported data in rugby union, whereby the mean time to return to play of 17–21 days was reported.[3,11] The rehabilitation that players

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 Mean (SD; range) time to return to full training in days and subsequent injury rate in different British Athletics Muscle Injury Classifications for hamstring muscle injuries.

BAMIC	*n*	Mean TTRFT in days: (SD; range)	Re-injury (same muscle)				Subsequent injury (same leg)				Subsequent injury (other leg)				Any recurrence
			Early	Mid-	Late	Total	Early	Mid-	Late	Total	Early	Mid-	Late	Total	Total
1a	6	21 (14; 8–56)	0%	17%	0%	17%	33%	17%	0%	14%	0%	0%	0%	0%	67%
1b	21	20.8 (13.6; 8–52)	10%	0%	0%	10%	10%	0%	0%	10%	0%	5%	14%	19%	38%
Grade 1	27	20.9 (14.0; 8–56)	7%	4%	0%	11%	15%	4%	0%	19%	0%	4%	11%	15%	44%
2a	7	15.7 (5.9; 9–24)	0%	0%	0%	0%	14%	14%	0%	29%	14%	0%	0%	14%	43%
2b	31	28.7 (21.2; 8–112)	10%	0%	3%	13%	6%	0%	6%	13%	10%	10%	0%	19%	45%
2c	11	43.7 (35.9; 9–128)	27%	0%	0%	27%	9%	0%	9%	18%	0%	18%	0%	18%	64%
Grade 2	49	30.2 (25.5; 8–128)	12%	0%	2%	14%	8%	2%	6%	16%	8%	10%	0%	18%	49%
3c	13	103 (40.1; 41–180)	0%	8%	0%	8%	8%	0%	0%	8%	15%	0%	8%	23%	38%
4c	2	105.5 (6.5; 99–112)	50%	0%	0%	50%	0%	0%	0%	0%	0%	0%	0%	0%	50%
Total	91	39.5 (38.2; 8–180)	10%	2%	1%	13%	10%	2%	3%	15%	7%	7%	4%	18%	46%

Abbreviations: BAMIC: British Athletics Muscle Injury Classification; TTRFT: time to return to full training.

received following HMI was underpinned by a bespoke, individualized, criteria-based approach which was overseen by the physiotherapist leading the rehabilitation of the specific injury case. In general, the approach of the club's physiotherapy and sports medicine team was to avoid pain during rehabilitation and running, which may explain some the longer times to return to full training. In addition, as exacerbations were considered prolonged index injuries, this may have had some impact on time to return to full training. Players took 3 times longer to return to full training from injuries with intramuscular tendon involvement versus injuries without intramuscular tendon involvement. Other authors have also reported significantly longer times to return to play for injuries with intramuscular tendon involvement.[16–18,21] Although Pollock et al.[16] reported a significantly longer time to return to full training following intramuscular tendon injury, these authors were specifically trialing a more conservative approach to rehabilitation with the primary objective being to mitigate the risk of subsequent injury. In their prospective double-blinded study, van der Made et al.[23] reported just an 8 day increase in time to return to play for injuries with intramuscular tendon involvement. This may suggest that if a clinician is not blinded to the MRI findings, their natural bias would be to treat intramuscular tendon injuries more cautiously. However, these findings warrant replication in other elite sports cohorts and across sports.

4.2 | Subsequent injury rates

We did not observe a statistically significant difference for the rate of subsequent injury when comparing HMIs with and without intramuscular tendon involvement. This aligns with the findings of McAuley et al.[19] and van der Made et al.[24] However, the rate of re-injury (13%) among our cohort was higher than that described by Pollock et al.[16] (2.9%) where the authors adopted the protocol described by Macdonald et al.[12] This suggests a similar approach to rehabilitation could potentially further reduce subsequent injuries.

4.3 | Disruption of T-Junction

We did not observe a statistically significant difference for the rate of subsequent injury when comparing femoris injuries with and without T-Junction involvement. Our study is the first to describe outcomes following T-Junction injuries in detail. It can be difficult to recognize data relating specifically to T-Junction injuries in other studies, due to contrasting reporting methods and study designs. However, Entwisle has reported

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

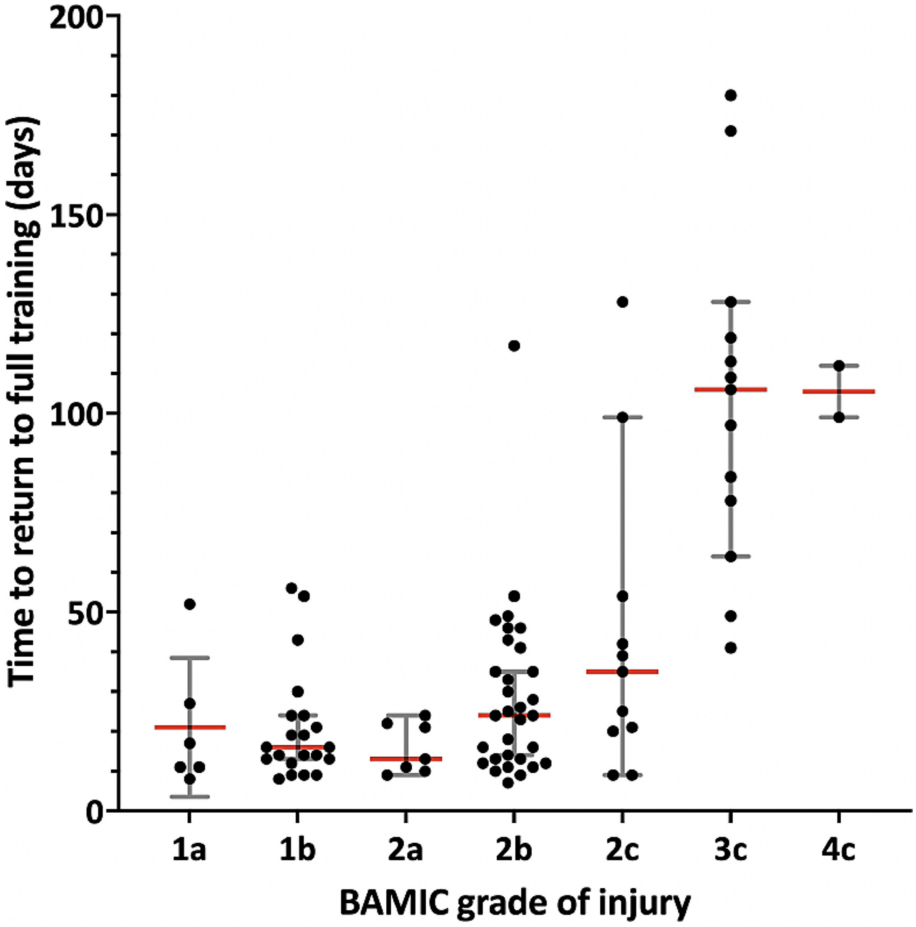

FIGURE 1 Time to return to full training for the different BAMIC grades of hamstring muscle injury.

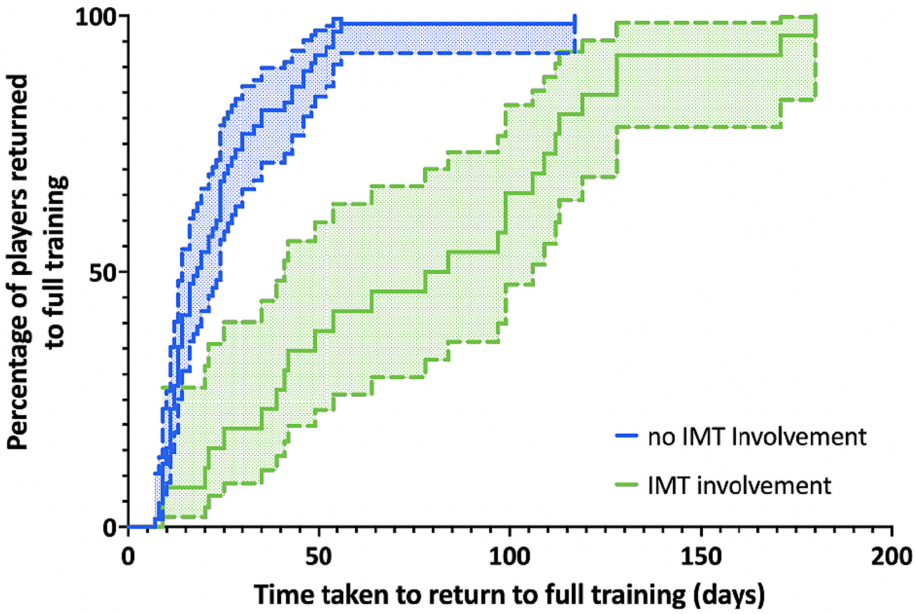

FIGURE 2 Kaplan Meier curve: Time to return to full training for hamstring muscle injuries with and without intramuscular tendon involvement.

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

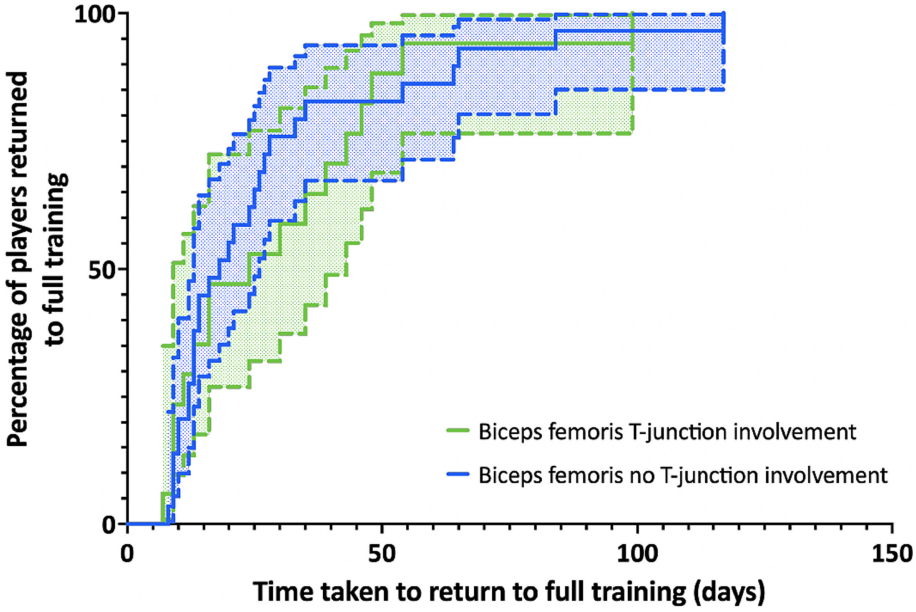

FIGURE 3 Kaplan Meier curve: Time to return to full training for biceps femoris injuries with and without T-junction involvement.

subsequent injury rates of 54% in a group primarily consisting of Australian Rules football players. Kayani et al.[30] anecdotally refers to subsequent injury rates off 55% in the authors experiences. Most re-injuries (5 of 9) at a Premier League football team involved the distal third of the biceps femoris, however the authors did not specify whether there was T-junction involvement.[18] Our data does not support the notion that biceps femoris injuries with T-junction involvement are a more serious type of HMI in professional rugby union players. Similarly, Pollock et al.[17] found no difference between distal injuries and proximal or central injuries for either time to return to full training or re-injury.

4.4 | Strengths and limitations

A strength of this study is the homogeneity of the participants—all were elite male professional rugby union players. This study is one of the largest to date to detail HMI outcomes using the BAMIC classification in professional sport. To our knowledge, this is the first study to do so in a professional rugby union setting. MRI scans were conducted using a 3T machine, offering superior resolution and signal-to-noise ratios compared to the 1.5T machines utilized in certain analogous studies.[18,23,24] This study provides greater insight into re-injury types than previous studies by detailing re-injuries to other hamstring muscles across both legs and for 12 months follow up.

However, there are a number of limitations to this work. All players were male. Although female players suffer less posterior thigh injuries than their male counterparts, HMIs are still a frequent cause of time loss in women's rugby union and this type of study would be worth replicating with data from the women's game.[31,32] The study was completed with data from single professional rugby union club in Ireland. This may bias the results toward the training practices of this club, and the biases and knowledge of the clubs sports medicine and physiotherapy practitioners. In addition, although our study describes a relatively large sample compared to other studies, a limited number of cases are available for each category (Table 2) making estimations of time to return to full training less accurate. This limitation can be applied to most of the existing literature in this area, and highlights the need for collaboration to increase the number of participants and thus, the accuracy of results.[33]

As this study was retrospective in nature, this provides potential sources of bias. MRI data were only available for injuries where it was deemed necessary to seek further clarification on diagnosis at the time of injury, and where the absence from sport was greater than 7 days. This results in selection bias against less severe injuries and may be reflected by the fact there are no Grade 0 HMIs included. It is possible that lower grade injuries such as Grade 1 HMIs are underrepresented in these findings as a result of these selection criteria. The study relies on retrospective inspection of the players' clinical notes via the club's EMRS. However, clinical notes primarily support diagnosis, treatment and rehabilitation of injuries rather than functioning as an injury surveillance system. In a small number cases, exact time to return to full training was not clear and consensus had to be reached between

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the authors as to the correct date. The ability to detect exacerbations (re-injuries during rehabilitation) was diminished by this approach and thus these were not reviewed as part of this study.

As in the recent study by McAleer et al.,[20] cases managed surgically were included for analysis. However, cases where surgery was delayed or required revision were excluded. Our study included the 2020 COVID-19 period – this was an atypical period of remote training, canceled fixtures and subsequent fixture congestion. This may have impacted upon the outcomes of the players who were injured during this period.

The authors acknowledge that a double-blinded prospective study design in a cohort of elite professional players would be the optimal study design. The players and clinicians in our study were not blinded to the MRI results, introducing a source of bias to the planning and management of rehabilitation. Evidence existed at the time to suggest that intramuscular tendon injuries may result in longer time to return to full training which may have affected the management of these injuries.[21] However, blinding the players and clinicians at the time of injury would not be feasible given the demands from players and coaches to optimize outcomes during the competitive season. In addition, the clinicians practices evolved in line with the increasing focus on these injuries within the published research. Accordingly, rehabilitation was inevitably adapted and evolved over time in line with the practitioners experiences and knowledge and reflective of the sporting context at the time (at both an individual or team level). However, this is an inevitable limitation of any data collected in an elite sporting environment.

5 | CONCLUSION

A retrospective analysis in professional men's rugby union revealed that players who sustain an acute HMI with intramuscular tendon disruption require a protracted time to return to full training. HMIs with either intramuscular tendon or T-junction disruption were not associated with an increased rate of re-injury/subsequent injury to the same limb. However, clinicians and players were not blinded to MRI findings at the time of injury, introducing a source of bias.

6 | PERSPECTIVE

We present the first study to describe the prevalence, time to return to full training and subsequent injury rates following intramuscular tendon and T-Junction injury of the hamstring in elite men's rugby union. To our knowledge, this is the first study to describe these outcome measures following T-Junction injury in any sport.

Our study found that rugby players who sustain a HMI with intramuscular tendon involvement should expect a protracted time to return to full training (three times longer than for HMIs not involving the intramuscular tendon). This is in keeping with the findings of other authors.[16–18,21,23] However, in another study where both the players and clinicians were blinded to MRI findings, van der Made et al.[23] found a difference in time to return to full training of just 8 days between HMIs with and without intramuscular tendon involvement. We acknowledge the difficulty of replicating this methodology in elite athletes. We did not find a difference in the rate of subsequent HMI at 12 months between HMIs with and without intramuscular tendon involvement.

Twenty per cent of the HMIs in this study involved the T-Junction of the biceps femoris, which is a higher proportion than previously reported (6%).[25] We did not find a significant difference in time to return to full training or subsequent injury when comparing HMIs involving the T-Junction to other biceps femoris injuries not involving the T-Junction.

AUTHORS CONTRIBUTIONS

The idea for the study was co-developed by FK, ED, GF, Darragh Curley, So'F, JC, UMP, and GDV. Data for the study was sourced by FK, So'F and DC. FK and So'F determined which MRI scans were potentially available for inclusion. The MRI Proforma (supplemental file 2) was developed by FK, So'F and GF, and approved by JC. JC carried out the retrospective reviews of the MRIs. FK and DC analyzed the club's electronic medical records (EMRS) to reach consensus on re-injuries and time to return to full training. JC carried out the retrospective reviews of the MRIs. ED and FK drafted the initial manuscript, with contributions from JC and So'F to the MRI section. This was edited and agreed upon by all authors.

ACKNOWLEDGMENTS

The authors would like to thank Dr Colin W. Fuller for assistance with re-injury type classification, and Professor John M. Ryan for assisting in the early stage of study feasibility design. We would also like to acknowledge all members of the Leinster Rugby medical team from 2015 to 2021 for their contribution to management of the injuries included. Open access funding provided by IReL.

FUNDING INFORMATION

Fearghal Kerin is the recipient of an Irish Research Council Employment-Based Postgraduate Scholarship (EBPPG/2018/100), with Professor Delahunt being the PI of the award.

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

CONFLICT OF INTEREST STATEMENT

No conflict of interest declared.

DATA AVAILABILITY STATEMENT

Deidentified data are available on reasonable request.

ORCID

Fearghal Kerin https://orcid.org/0000-0002-7300-4120

REFERENCES

1. Ekstrand J, Hagglund M, Walden M. Epidemiology of muscle injuries in professional football (soccer). *Am J Sports Med*. 2011;39(6):1226-1232. doi:10.1177/0363546510395879
2. Gabbe BJ, Bennell KL, Finch CF, Wajswelner H, Orchard JW. Predictors of hamstring injury at the elite level of Australian football. *Scand J Med Sci Sports*. 2006;16(1):7-13. doi:10.1111/j.1600-0838.2005.00441.x
3. Brooks JH, Fuller CW, Kemp SP, Reddin DB. Incidence, risk, and prevention of hamstring muscle injuries in professional rugby union. *Am J Sports Med*. 2006;34(8):1297-1306. doi:10.1177/0363546505286022
4. Dalton SL, Kerr ZY, Dompier TP. Epidemiology of hamstring strains in 25 NCAA sports in the 2009-2010 to 2013-2014 academic years. *Am J Sports Med*. 2015;43(11):2671-2679. doi:10.1177/0363546515599631
5. Kenneally-Dabrowski C, Serpell BG, Spratford W, et al. A retrospective analysis of hamstring injuries in elite rugby athletes: more severe injuries are likely to occur at the distal myofascial junction. *Phys Ther Sport*. 2019;38:192-198. doi:10.1016/j.ptsp.2019.05.009
6. Kerin F, Farrell G, Tierney P, McCarthy Persson U, De Vito G, Delahunt E. Its not all about sprinting: mechanisms of acute hamstring strain injuries in professional male rugby union-a systematic visual video analysis. *Br J Sports Med*. 2022;56(11):608-615. doi:10.1136/bjsports-2021-104171
7. Fuller CW, Taylor A, Douglas M, Raftery M. Rugby world cup 2019 injury surveillance study. *S Afr J Sports Med*. 2020;32:v32i1a8062. doi:10.17159/2078-516X/2020/v32i1a8062
8. de Visser HM, Reijman M, Heijboer MP, Bos PK. Risk factors of recurrent hamstring injuries: a systematic review. *Br J Sports Med*. 2012;46(2):124-130. doi:10.1136/bjsports-2011-090317
9. Koulouris G, Connell DA, Brukner P, Schneider-Kolsky M. Magnetic resonance imaging parameters for assessing risk of recurrent hamstring injuries in elite athletes. *Am J Sports Med*. 2007;35(9):1500-1506. doi:10.1177/0363546507301258
10. Wangensteen A, Tol JL, Witvrouw E, et al. Hamstring Reinjuries occur at the same location and early after return to sport: a descriptive study of MRI-confirmed Reinjuries. *Am J Sports Med*. 2016;44(8):2112-2121. doi:10.1177/0363546516646086
11. Bourne MN, Opar DA, Williams MD, Shield AJ. Eccentric knee flexor strength and risk of hamstring injuries in Rugby union: a prospective study. *Am J Sports Med*. 2015;43(11):2663-2670. doi:10.1177/0363546515599633
12. Macdonald B, McAleer S, Kelly S, Chakraverty R, Johnston M, Pollock N. Hamstring rehabilitation in elite track and field athletes: applying the British athletics muscle injury classification in clinical practice. *Br J Sports Med*. 2019;53(23):1464-1473. doi:10.1136/bjsports-2017-098971
13. Reurink G, Brilman EG, de Vos RJ, et al. Magnetic resonance imaging in acute hamstring injury: can we provide a return to play prognosis? *Sports Med*. 2015;45(1):133-146. doi:10.1007/s40279-014-0243-1
14. Wangensteen A, Guermazi A, Tol JL, et al. New MRI muscle classification systems and associations with return to sport after acute hamstring injuries: a prospective study. *Eur Radiol*. 2018;28(8):3532-3541. doi:10.1007/s00330-017-5125-0
15. Pollock N, James SL, Lee JC, Chakraverty R. British athletics muscle injury classification: a new grading system. *Br J Sports Med*. 2014;48(18):1347-1351. doi:10.1136/bjsports-2013-093302
16. Pollock N, Kelly S, Lee J, et al. A 4-year study of hamstring injury outcomes in elite track and field using the British athletics rehabilitation approach. *Br J Sports Med*. 2022;56(5):257-263. doi:10.1136/bjsports-2020-103791
17. Pollock N, Patel A, Chakraverty J, Suokas A, James SL, Chakraverty R. Time to return to full training is delayed and recurrence rate is higher in intratendinous ('c') acute hamstring injury in elite track and field athletes: clinical application of the British athletics muscle injury classification. *Br J Sports Med*. 2016;50(5):305-310. doi:10.1136/bjsports-2015-094657
18. Shamji R, James SLJ, Botchu R, Khurniawan KA, Bhogal G, Rushton A. Association of the British Athletic Muscle Injury Classification and anatomic location with return to full training and reinjury following hamstring injury in elite football. *BMJ Open Sport Exerc Med*. 2021;7(2):e001010. doi:10.1136/bmjsem-2020-001010
19. McAuley S, Dobbin N, Morgan C, Goodwin PC. Predictors of time to return to play and re-injury following hamstring injury with and without intramuscular tendon involvement in adult professional footballers: a retrospective cohort study. *J Sci Med Sport*. 2022;25(3):216-221.
20. McAleer S, Macdonald B, Lee J, et al. Time to return to full training and recurrence of rectus femoris injuries in elite track and field athletes 2010-2019; a 9-year study using the British athletics muscle injury classification. *Scand J Med Sci Sports*. 2022;32(7):1109-1118. doi:10.1111/sms.14160
21. Comin J, Malliaras P, Baquie P, Barbour T, Connell D. Return to competitive play after hamstring injuries involving disruption of the central tendon. *Am J Sports Med*. 2013;41(1):111-115. doi:10.1177/0363546512463679
22. Eggleston L, McMeniman M, Engstrom C. High-grade intramuscular tendon disruption in acute hamstring injury and return to play in Australian football players. *Scand J Med Sci Sports*. 2020;30(6):1073-1082. doi:10.1111/sms.13642
23. van der Made AD, Almusa E, Whiteley R, et al. Intramuscular tendon involvement on MRI has limited value for predicting time to return to play following acute hamstring injury. *Br J Sports Med*. 2018;52(2):83-88. doi:10.1136/bjsports-2017-097659
24. van der Made AD, Almusa E, Reurink G, et al. Intramuscular tendon injury is not associated with an increased hamstring reinjury rate within 12 months after return to play. *Br J Sports Med*. 2018;52(19):1261-1266. doi:10.1136/bjsports-2017-098725
25. Entwisle T, Ling Y, Splatt A, Brukner P, Connell D. Distal Musculotendinous T junction injuries of the

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

biceps Femoris: an MRI case review. *Orthop J Sports Med*. 2017;5(7):2325967117714998. doi:10.1177/2325967117714998
26. Fuller CW, Bahr R, Dick RW, Meeuwisse WH. A framework for recording recurrences, reinjuries, and exacerbations in injury surveillance. *Clin J Sport Med*. 2007;17(3):197-200. doi:10.1097/JSM.0b013e3180471b89
27. Mansournia MA, Collins GS, Nielsen RO, et al. A CHecklist for statistical assessment of medical papers (the CHAMP statement): explanation and elaboration. *Br J Sports Med*. 2021;55(18):1009-1017. doi:10.1136/bjsports-2020-103652
28. Brukner P, Connell D. 'Serious thigh muscle strains': beware the intramuscular tendon which. *Br J Sports Med*. 2016;50(4):205-208.
29. Schuermans J, Van Tiggelen D, Danneels L, Witvrouw E. Susceptibility to hamstring injuries in soccer: a prospective study using muscle functional magnetic resonance imaging. *Am J Sports Med*. 2016;44(5):1276-1285. doi:10.1177/0363546515626538
30. Kayani B, Ayuob A, Begum F, Singh S, Haddad FS. Surgical repair of distal Musculotendinous T junction injuries of the biceps Femoris. *Am J Sports Med*. 2020;48(10):2456-2464. doi:10.1177/0363546520938679
31. Taylor AE, Fuller CW, Molloy MG. Injury surveillance during the 2010 IRB Women's Rugby world cup. *Br J Sports Med*. 2011;45(15):1243-1245. doi:10.1136/bjsports-2011-090024
32. Fuller CW, Taylor A. Eight-season epidemiological study of match injuries in women's international rugby sevens. *J Sports Sci*. 2021;39(8):865-874. doi:10.1080/02640414.2020.1850616
33. van Dyk N, van der Made AD, Timmins RG, Opar DA, Tol JL. There is strength in numbers for muscle injuries: it is time to establish an international collaborative registry. *Br J Sports Med*. 2018;52:1228-1229.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Kerin F, O'Flanagan S, Coyle J, et al. Are all hamstring injuries equal? A retrospective analysis of time to return to full training following BAMIC type 'c' and T-junction injuries in professional men's rugby union. *Scand J Med Sci Sports*. 2024;34:e14586. doi:10.1111/sms.14586

16000838, 2024, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.14586, Wiley Online Library on [22/02/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License